EAST WEST UNIVERSITY

LAB PROJECT

"Bench Vice Sliding Jaw"

Department of CSE

Semester: Spring 2026

Course Code: CSE 200

Course Title: Engineering Drawing

Section: 09

Submitted By:	Submitted To:
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Department: CSE	Department: CSE

Date of Submission: 14/04/2026

Project Title:

Design and Modeling of a Bench Vice Sliding Jaw

Project Description:

Overview: This project involves the precise 3D modeling and 2D engineering drafting of a **Sliding Jaw**, a critical component of a standard Bench Vice. The object was selected for its geometric complexity, requiring a combination of advanced AutoCAD operations to demonstrate technical proficiency in Computer-Aided Design.

About: A bench vice sliding jaw (or movable jaw) is the dynamic, moving component of a vice that shifts forward and backward to clamp or release workpieces. It is attached to a slide that moves along the vice body, operated by a threaded screw and handle to exert pressure against the fixed jaw

Technical Details: The model was developed using a bottom-up approach, starting with a 2D profile based on standard mechanical dimensions. The primary features of this design include:

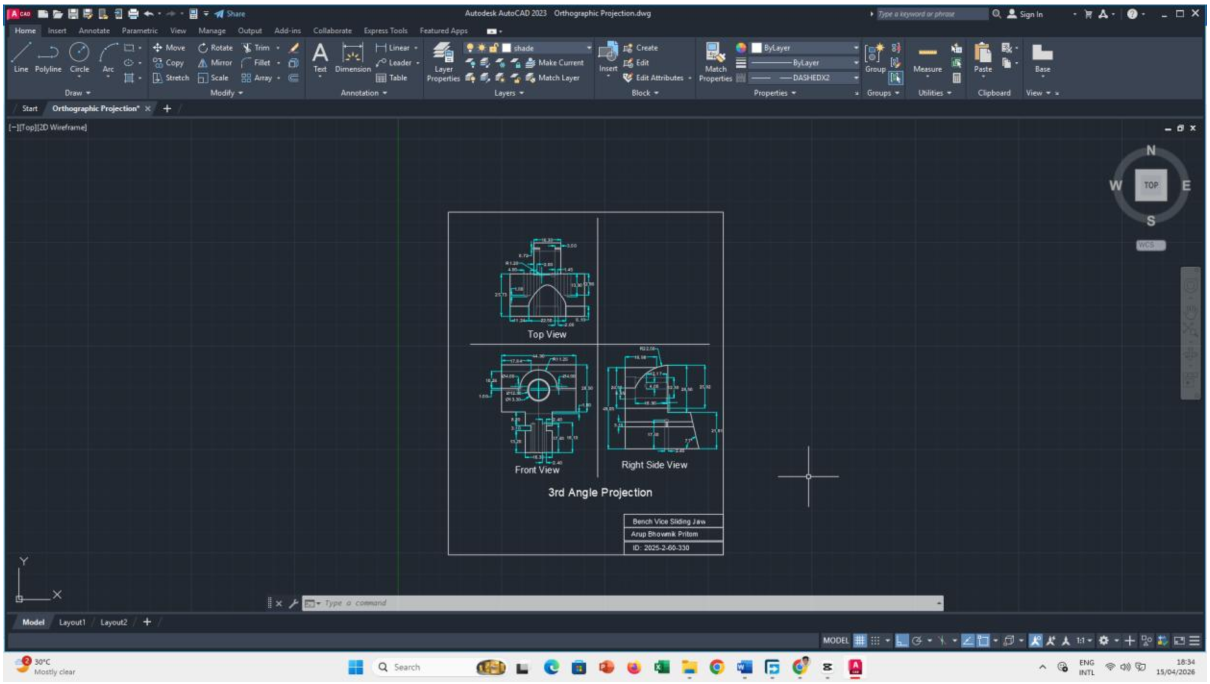
- **3D Solid Modeling:** Created using EXTRUDE and UNION operations to ensure a manifold solid structure.
- **Boolean Operations:** Integrated slots and screw housing holes were precisely crafted using the SUBTRACT command to mimic real-world machining processes.
- **Geometric Precision:** The design incorporates curved profiles (fillets) and angular surfaces to ensure realistic mechanical functionality.

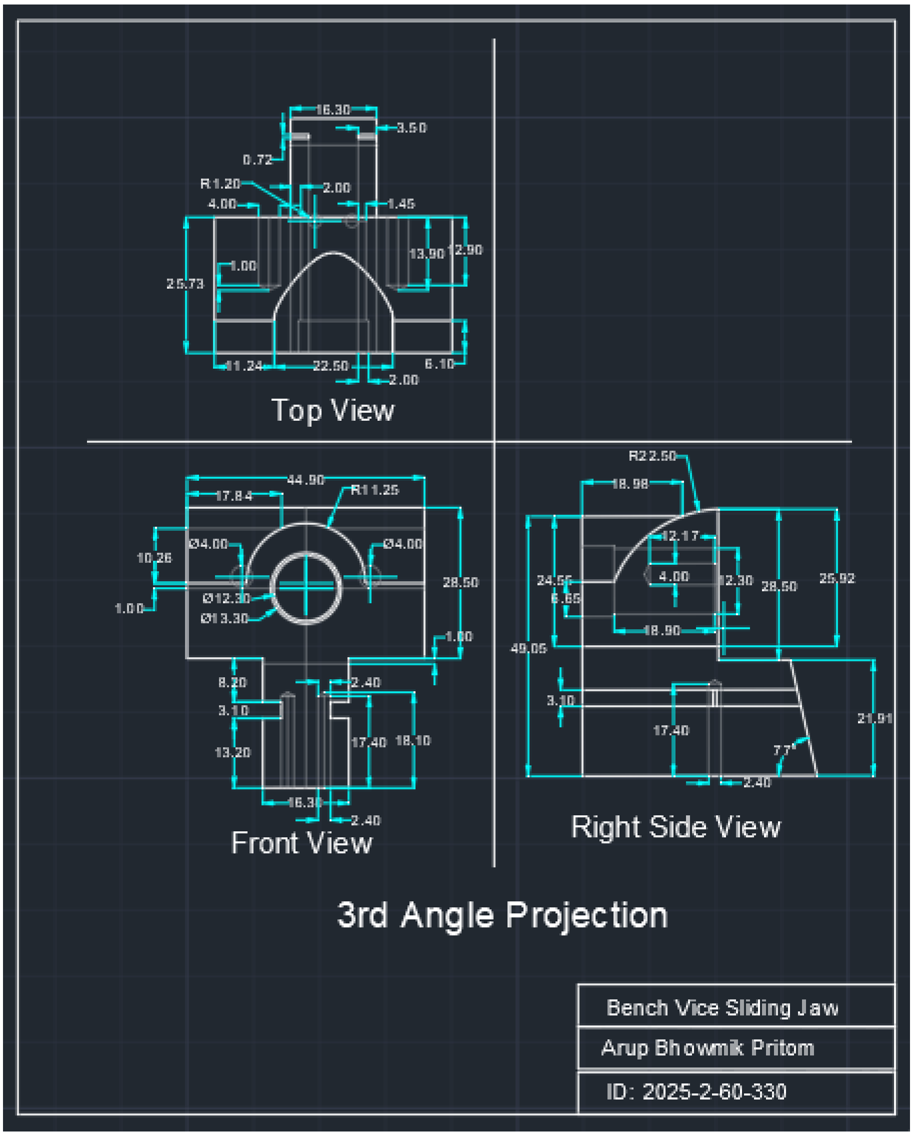
Drafting & Projections: To provide a comprehensive engineering view, the project includes:

- **Orthographic Projections:** I used **3rd Angle Projection**, showcasing the Front, Top, and Side views with accurate hidden line details.
- **Isometric View:** 2D isometric representation to provide a clear spatial understanding of the component.
- **3D Modeling:** high-fidelity 3D solid model representing the final physical form of the Bench Vice Sliding Jaw, showcasing its structural integrity and mechanical design.
- **Annotations:** Comprehensive dimensioning and labeling to ensure the design meets industrial documentation standards.

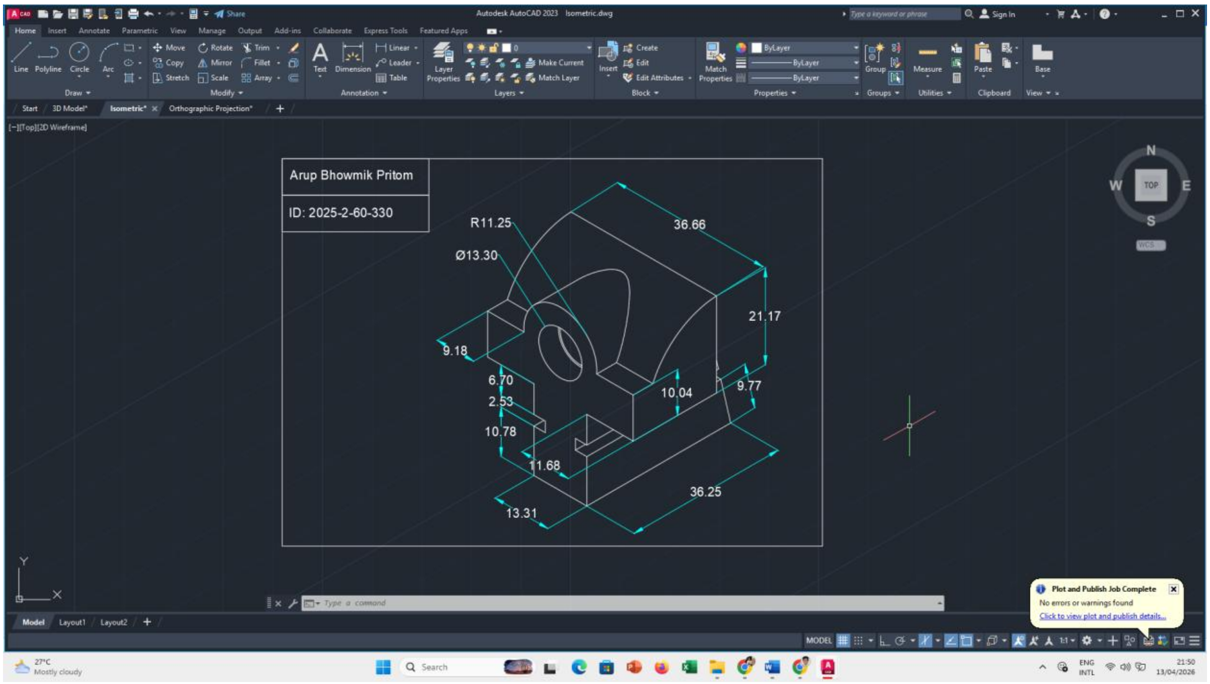
Software Used: AutoCAD 2023

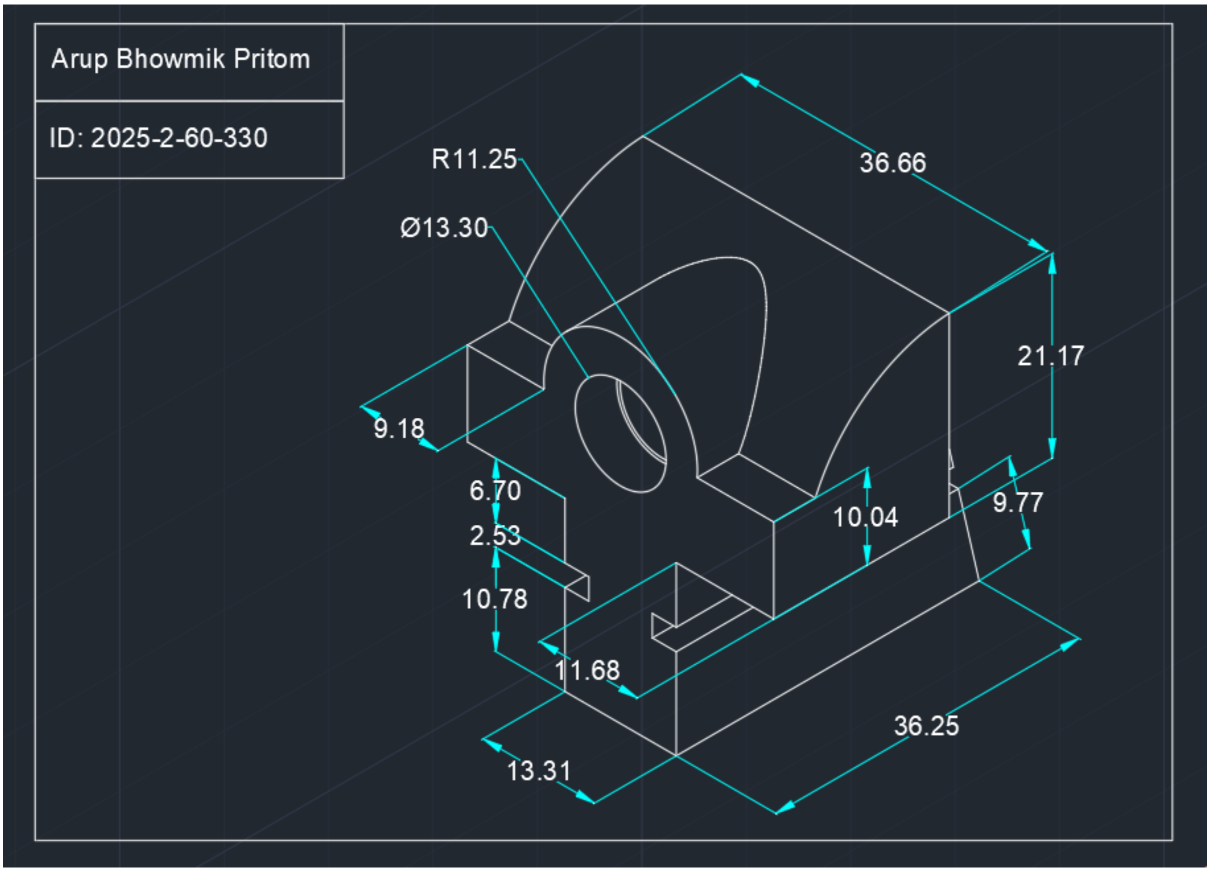
Orthographic Projection (3^{rd} Angle) :



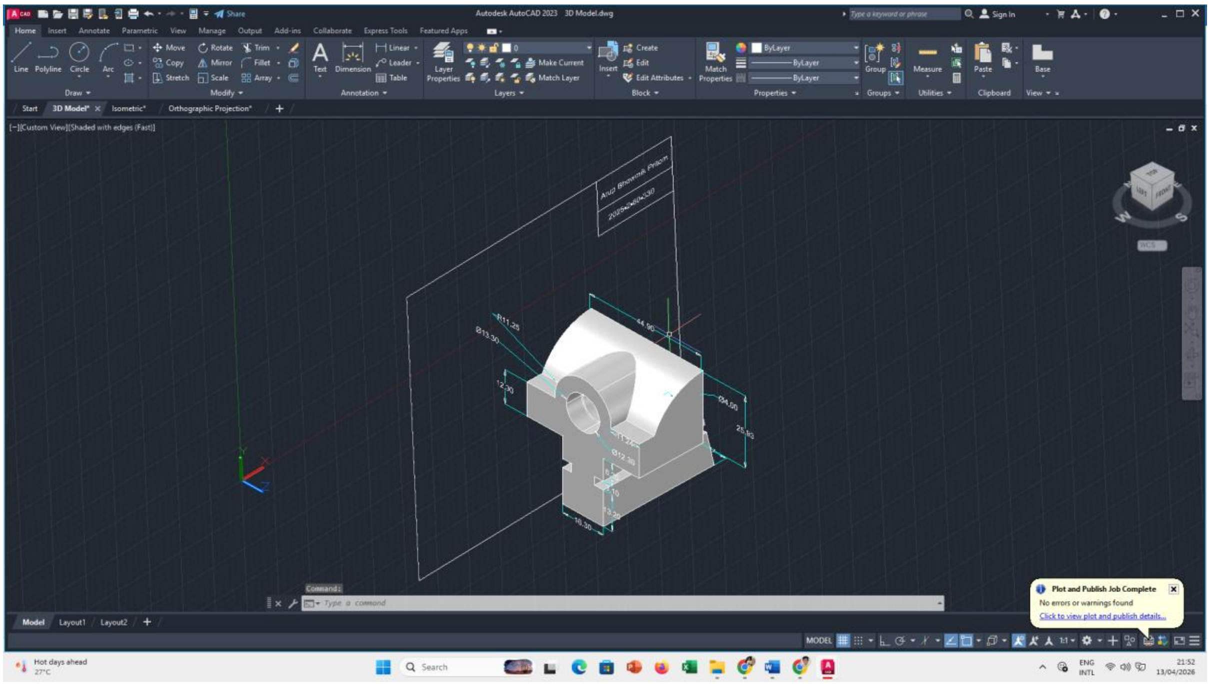


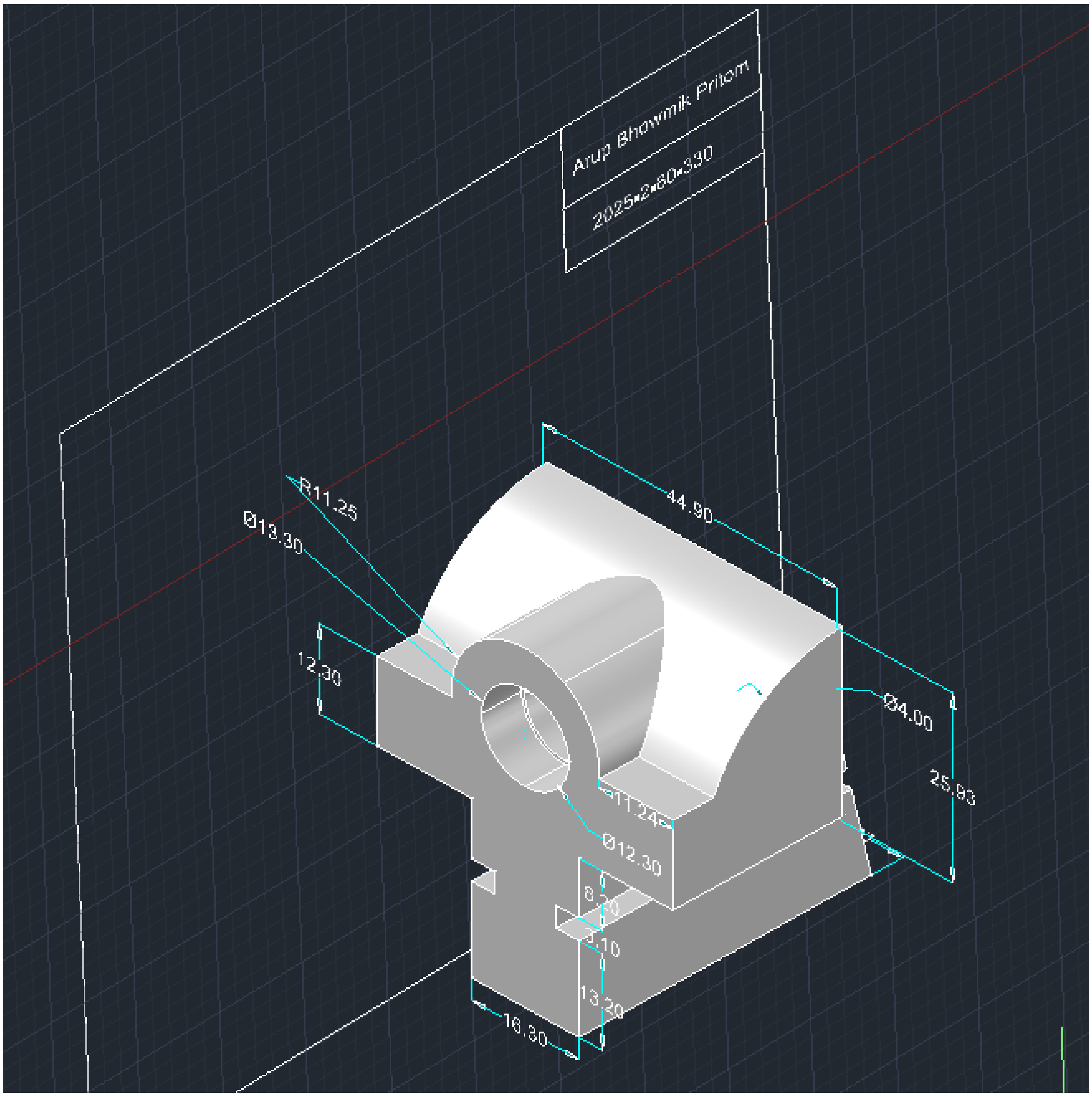
Isometric View :





3D Model :

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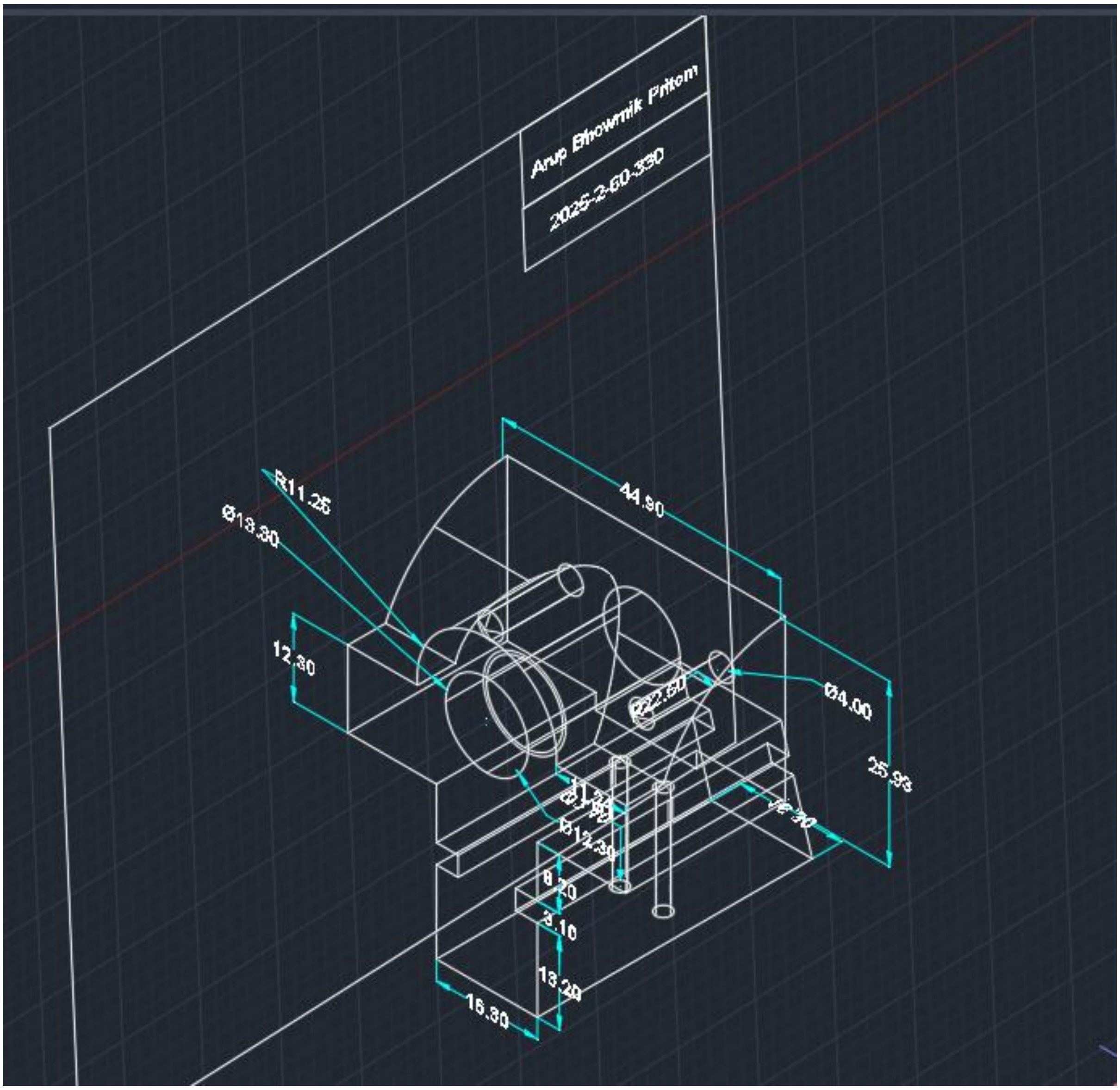

Thank You Sir